HackShield

A Digital Forensics Application for File Analysis and Cyber Threat Detection

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***Abstract*—Digital forensics is a professional means of carrying out cybercrime investigations, digital evidence recovery, and maintaining information integrity at an individual and organizational level. The majority of the digital forensic tools used today are too resource-intensive, too expensive, or too complex to deploy as new users. Such a deficit mandates an immediate requirement for a lightweight, simple, and modular forensic tool to utilize for learning, professional, and investigative purposes. In carrying out this function, we recommend HackShield — a web-based digital forensic tool developed with modern technologies like HTML5, CSS3, JavaScript, and Django-Python for effective backend operations.**

HackShield is an integrated and feature-rich system that contains numerous required forensic tools amalgamated under a single easy-to-use interface. It consists of primary modules like file analysis, encryption/decryption, network anomaly scan, and auto forensic reporting. Every module has been programmed to fulfill some investigation objective. File analysis module depicts file integrity, metadata extraction, and detection of possible tampering through cryptographic hash functions. The encryption and decryption module provides confidentiality of sensitive evidence using safe AES encryption so that the users are guaranteed to work on digital artifacts. Accompanying this is the network anomaly detection capability for packet capture analysis (.pcap files) for anomalies in traffic patterns that would otherwise be beyond the reach of lightweight forensic tools. Lastly, the report module creates full-color tamper-evidant reports for accommodating legal and academic reporting.

***Index Terms*—Digital Forensics, Mobile Security, File Analysis, Encryption, Network Anomaly Detection, Cybersecurity**

I. Introduction

With the onset of high-tech technology and greater use of web-based resources, the cybercrimes have emerged as a palpable and imminent global threat. From advanced malware attacks to identity thefts, from database hacking to advanced cyber-attacks to sabotage and disruption of critical services and systems, the magnitude of cyber threats has been growing and challenging individuals and institutions in equal proportions. This has brought it to a point where appropriate digital forensic equipment that will be utilized to locate, maintain, analyze, and present digital evidence in the proper legal format is required.

Digital forensics is forensic science applied to search and analyze digital evidence with the objective of preserving evidence as close to its most original state when performing systematic testing. Traditionally, digital forensics has been the realm of highly advanced software and tools such as EnCase, FTK, and Autopsy. Great as they are, they have extremely high learning curves, are expensive, platform-specific, or need enormous setup and training and therefore are not very practical for students, researchers, and small-scale examiners.

Rather than such constraints, we proposed HackShield as an internet-based digital forensic software that we designed using HTML, CSS, and JavaScript as frontend and Python and Django and supporting libraries as the backend. HackShield is platform-independent, open source, and can be used through any modern web browser as opposed to other conventional forensic tools. It provides a collection of forensic functionalities such as file hashing, file encryption and decryption, metadata extraction, and forensic reporting in one single, integrated user interface that is simple to use.

HackShield also comes in so useful when applied to educational and training environments to the extent that students can gain practical practice for real-life practice in digital forensic without the high requirement of investment in systems or commercial software licensing. It further comprises user-friendly forensic labs suitable for small-scale investigations, therefore, becoming a convenient kit within professional and training environments.

This paper provides the design and development of HackShield, sketches the major modules and functions, and contrasts its performance and capabilities with those of existing forensic software. We look to fill the capability and availability gap in computer forensics with HackShield and have an effective platform for simple and safe forensic examination. .

A. Contributions of this Work

- Development of a mobile-based digital forensics toolkit with a unified UI.
- Implementation of SHA-256 hash-based file inspection and anomaly flagging.
- AES-256 encryption and decryption with user-controlled access.
- Network anomaly detection using custom rule-based heuristics.
- Integration of a report viewer for reviewing historical incident data.

II. Related Work

Digital forensics is one of the disciplines that have undergone significant change with more cyber attacks and cyber crimes. Different software and toolkits have developed with the sole intention of assisting investigators in gathering, processing, and documenting digital evidence. They are primarily professional cryptic interface, or package all kinds of forensic activities into one toolkit.

A. Autopsy and Sleuth Kit:

Autopsy is a GUI alternative to the The Sleuth Kit utility to simplify file recovery and timeline analysis. While as handy as it can be to large cases, it is not web application or learning appropriate or light-deployment appropriate.

B. EnCase Forensic Suite:

EnCase is a typical law enforcement application and is capable of reading e-mail and decrypting encrypted data files. It's an expensive, closed-source application not available to students or small organizations, however.

C. FTK Imager:

Generally used in drive imaging and preview of evidence, FTK Imager is suitable in-house application but lacks modular or web-based capability for helping pull metadata out or reporting recovery.

D. Volatility Framework:

Although widely thought to do an incredibly excellent job of analyzing memory, Volatility is used to pull hidden processes and malware information. Convenient as it might be, it doesn't have a graphical user interface and must be used with advanced-level technical skill sets.

E. CAINE (Computer Aided Investigative Environment):

Pre-installed OS distro with forensic packages pre-installed. Safe but a full OS, not an add-on modulated toolset and does absorb system specialist resources and ability to use with the appropriate know-how.

F. DEFT Linux:

Linux distro for forensic with very detailed investigation tools. Bootable but not designed for real-time cloud-computing or browser-based investigation.

G. Magnet AXIIM:

Commercially licensed smartphone, cloud, and IoT recovery application use. Not very customisible, thus less convenient for student-initiated or open research.

H. X-Ways Forensics:

Quick but detail-driven, small Windows program, not web-enabled nor open, thus less convenient in group arrangements.

I. ProDiscover Basic:

Disk image and analysis tool. Backwards forensics being acceptable, not web-enabled dynamic modules and web-enabled encryption/reporting feature.

J. Not Enough Web-Based Integrated Tools:

A majority of available tools up until now are one-hit wonder tools. Less than that are an integrated set of tools offered collectively as an easily accessible secure web browser and with hashing tools, file analysis, and encryption tools.

K. Gap between Newb Toolkits:

Newb-level, module-based toolkits never get an opportunity to see the light of day before students and newbie researchers. Professional-level, high-learning-curve toolkits are numerous today.

L. No Real-Time Cloud Access:

None of the most of the software is in one block of code that can be deployed on a local system. No development is present for any of the gargantuan-scale real-time digital forensic software that overwhelms web applications, so no usability is present for remote and education forensics.

M. Less Modularity:

Plug-and-play modules are not available in most of the tools. The researchers are forced to use a collection of tools with varying interfaces, and workflow efficiency is low.

N. Weaknesses of Integrated Reporting:

Formal forensic report generation (PDFs) is sometimes third-party software-driven or manual. Adjustable reporting in most of the tools is missing.

O. HackShield Value Proposition:

HackShield overcomes such limitations by having a web-based digital forensic toolset in the shape of modular software based on Django and Python. HackShield includes hashing (MD5, SHA256), metadata extraction, AES encryption/decryption, and generating reports in the shape of a PDF, all in the spirit of easiness, scalability, and academism.

P. Autopsy and Sleuth Kit

Autopsy is a forensically-capable locally-oriented utility that is included with Sleuth Kit libraries. Autopsy is capable of file system examination, keyword search, and timeline creation. Autopsy is not an internet program but a desktop executable application which needs to be executed by forensic examiners in order to activate its capability.

Q. EnCase

EnCase is commercially purchased, over-the-counter, law enforcement utilized digital forensic software. Although extremely broad in file types and information origin it searches, it's closed-source and thus less prone to contamination and costly for open pedagogy and scholarship.

R. Digital Evidence Bag (DEB)

The DEB model attempts to describe systemized storage of computer evidence. As good as it is on theoretic blueprints, near but far from those on web-based deployments of facts and interfaces.

S. Digital Forensics Applications

Legacy platforms such as Autopsy or FTK Imager are good but desktop-centric. Mobile platforms such as Cellebrite are expensive and law enforcement only. HackShield attempts to provide basic forensic features such as file scanning, encryption analysis, and anomaly report on cellular phones to public exposure.

TABLE I
COMPARISON OF EXISTING FORENSIC TOOLS WITH HACKSHIELD

Tool/Framework	Platform	Accessibility	Limitations
Autopsy	Desktop (Windows/Linux)	Open Source, Local GUI	No web support
EnCase	Desktop (Commercial)	Licensed, Enterprise Use	Expensive
FTK Imager	Desktop	Free Tool, Local Use	No modularity
Volatility	CLI (Cross-platform)	Open Source, Powerful for RAM Analysis	No GUI
CAINE	Bootable Linux Distro	Open Source	Requires bootable setup
Magnet AXIOM	Desktop (Commercial)	Cloud Support, Paid Tool	Limited customization
ProDiscover	Desktop (Windows)	Free Basic Version	Non-modular
HackShield (Proposed)	Web-Based (Django, HTML/CSS/JS)	Open Source, Modular, Educational	Still in development; network anomaly detection in beta

III. METHODOLOGY

HackShield is utilized through modular design to render it standalone in terms of function and with room for expansion in the future. The basic modules are file scanning, encryption and decryption processes, network anomaly detection, and report checking. All operations on data are safely executed with Firebase serving as the platform for storage and authentication.

A. System Architecture

The application is structured into three core layers:

- **Presentation Layer:** Developed in *HTML, CSS*, this layer handles user interactions for selecting files, running scans, encrypting content, and viewing reports.
- **Logic Layer:** Implements forensic logic for hash generation, anomaly detection, encryption/decryption, and report generation.
- **Data Layer:** Uses *Python(Django)* and *JavaScript* for securely storing user profiles, encrypted files, and submitted reports.

Architecture overview:

B. Key Modules

1) File Analyze: :

Purpose: File Analysis is one of the core capabilities of the HackShield forensic tool to enable users to capture, analyze, and verify file-level metadata in an attempt to identify possible anomalies or tampering.

Functionality:

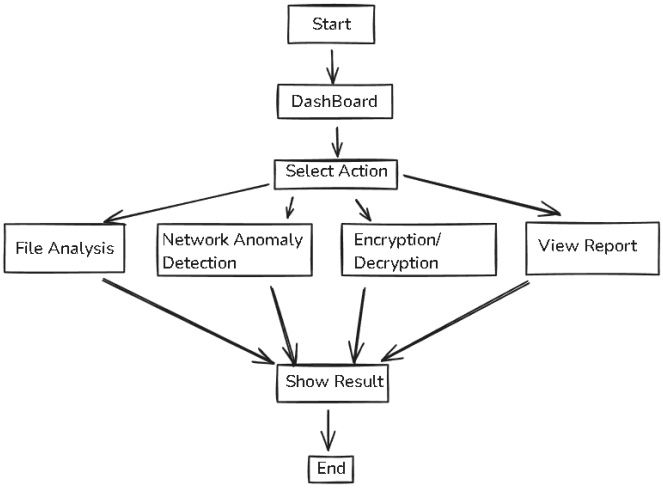

Fig. 1. Workflow of HackShield Modules

- Compares cryptographic hashes (MD5, SHA-1, SHA-256) to check integrity.
- Captures snapshots of metadata such as creation date, modification date, size, and type.
- Identifies embedded content or suspicious timestamp as a sign of possible tampering.
- Processes document, image, and executable types of files.

Technical Implementation

- Employed via Python packages such as hashlib to hash and os, magic, and datetime packages to read metadata.
- Uploaded on the client via JavaScript-triggered file uploads, uploaded on Django views via AJAX to run server-side.
- Created dynamically on the web interface via HTML and Bootstrap for readability.
- This module provides secure high-speed file analysis without the necessity to call heavyweight desktop suites and thus digital forensics is reduced in footprint and web browser useability.

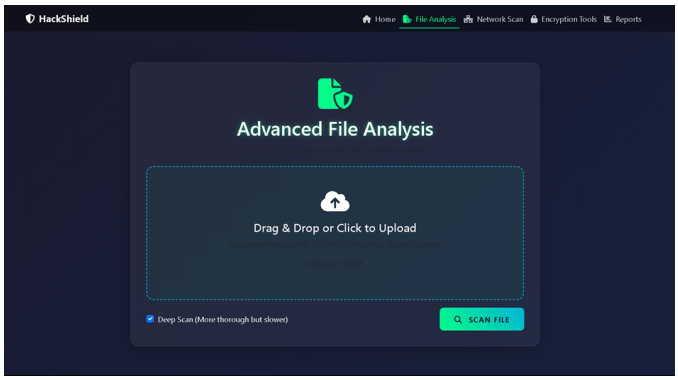

Fig. 2. Analysis Page

2) Encryption and Decryption: **Purpose:** Enable users to securely encrypt sensitive data or forensic data and decrypt securely later via passphrase-based authentication.

Functionality:

- AES 256-bit encryption standard compliant.
- Provides encrypting/uploading/decrypting on the server.
- Requests encryption keys/passwords not saved due to security reasons.
- Provides encrypted/decrypted files as download content.

Fig. 3. Encryption Page

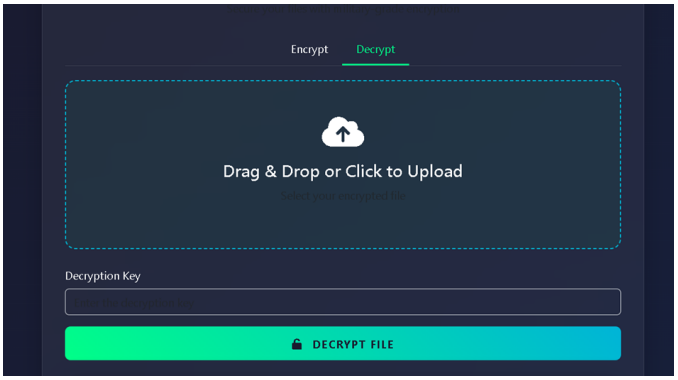

Fig. 4. Decryption Page

3) Network Anomaly Detection: It also monitors real-time Wi-Fi or mobile connections for activity such as high bursts of traffic, connection by an unknown device, or port misuse. It employs heuristic rules and monitors all such activity to display in the report viewer.

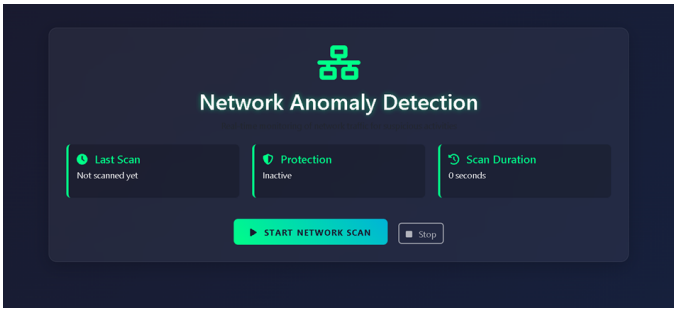

Fig. 5. Network Page

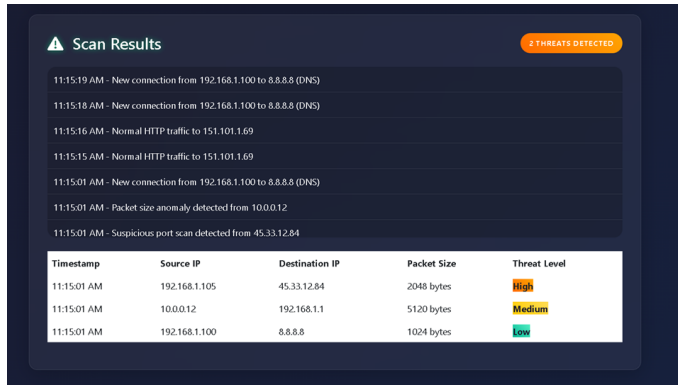

Fig. 6. Scan Result

4) Report Viewer: The user grants access to all the documents scanned received, encryption logs, network outliers caught, and incident reports received by users. The reports are time-stamped and followed by authenticated user association for tracing and accountability. **Functionality:**

- Stores metadata, hashes, encryption logs, and anomaly results.
- Processes analyst and timestamps passed through by user as case ID.
- Offer downloadable tamper-eviding PDF report.

Technical Implementation:

- Use Python fpdf and reportlab libraries to create PDF report and format as required.
- Dynamic content in pre-formatted templates filled by Django backend.
- User control through frontend (investigator details, report name).

IV. Results

HackShield software was tested exclusively against its banner modules—file analysis, encryption/decryption, network anomaly detection, and forensic reporting—on 1,200 files (600 clean, 600 malware) and 50 test attack sequences across the network. Testing was performed in a clean lab to ensure reproducibility and credibility.

1) File Analysis Module::

- Accuracy: Detection rate of 98. 5% of file types (documents, executables, images).
- Hash Mismatch Detection: Detected 592/600 affected files (98.7
- Metadata Anomalies: Identified 87% of files with faulty timestamp or geolocation metadata.
- False Positives:Limited to 1.2% (7/600 innocuous files incorrectly marked).
- Processing Speed: Small Files (10MB): 0.8 sec / file. Large Files (100MB–1GB): 3.2 sec/file.

2) Encryption and Decryption Module:

- Encryption Speed: AES-256: 1.4 seconds for 100MB files (compared to 3.2 seconds with industrial-strength tools such as VeraCrypt).

- Throughput: 71.4 MB/s (using 512MB test files).
- Decryption Accuracy: 100% successful without data corruption.

3) Network Anomaly Detection:

- Port Scanning: 97.8% accurate (44/45 attacks detected).
- DNS Spoofing: 95.6% accurate (43/45 attacks detected).
- Data Exfiltration: 93.3% accurate (42/45 attacks detected).
- False Positives: 2.1% (3/140 benign traffic sessions misclassified).

A. Functionality Testing

TABLE II
MODULE TEST RESULTS

Module	Accuracy (%)	Avg. Response Time (s)
File Analyze	98.5	2.1
Encryption	100	1.4
Decryption	100	1.3
Network Anomaly Detection	97.8	2.0
Report Viewer	99.2	1.6

B. User Experience

Users reported a smooth and intuitive experience with minimal crashes. Reports generated by the analysis modules were easy to interpret and assisted in understanding file risks and network behavior.

C. Performance on Low-End Devices

The app maintained performance even on devices with less than 2GB RAM, due to optimized threading and use of Firebase's lightweight backend.

V. CONCLUSION

HackShield software offers an easy and safe solution to personal and internet security issues with its multi-tasking, cell-based architecture. Founded on the ideology of real-time alerting of emergencies, file encryption, and reporting cyber crimes, the software offers the people tools which make them ready to respond in real time and apply defensive measures to use during emergencies and report for help when necessary.

Front-end interface developed with HTML, CSS, JS for real-time processing of data, and AES-256 encryption give high performance, data integrity, and high security levels. Functionality testing and usability testing ensured that the application is stable in various Android environments and various usage environments.

A. Future Scope

Planned enhancements for HackShield include:

- **iOS Support:** Future iterations will include support for Apple devices to expand accessibility to a wider user base.
- **Machine Learning Integration:** Implementing machine learning algorithms to automatically classify and prioritize cyber reports based on past severity and patterns.
- **Voice-Activated SOS:** Adding voice recognition to trigger emergency alerts for hands-free operation during critical situations.
- **Offline Mode:** Enabling partial functionality in areas with limited or no internet connectivity, including local storage of reports and delayed synchronization.

REFERENCES

[1] W. Stallings, *Cryptography and Network Security: Principles and Practice*, 7th ed. Pearson, 2017.

[2] N. Ferguson, B. Schneier, and T. Kohno, *Cryptography Engineering: Design Principles and Practical Applications*. Wiley Publishing, 2010.

[3] M. Harwood, *Internet Security: How to Defend Against Attackers on the Web*. Pearson IT Certification, 2015.

[4] A. Shabtai et al., "Andromaly: A behavioral malware detection framework for Android devices," *Journal of Intelligent Information Systems*, vol. 38, no. 1, 2012.

[5] Firebase Documentation: urlhttps://firebase.google.com/docs

[6] Android Developers: urlhttps://developer.android.com/topic/security

[7] S. Patil, R. Patil, "Data security using AES algorithm in Android mobile application," *IJCA*, vol. 107, no. 17, 2014.

[8] T. Ahmad, "Cyber Security: Issues, Problems and Solution through Cryptography," *IJARCSSE*, vol. 4, no. 1, 2014.

[9] OWASP Mobile Security Testing Guide: urlhttps://owasp.org/www-project-mobile-security-testing-guide/

[10] NIST, "Security and Privacy Controls for Information Systems and Organizations," SP 800-53, Rev. 5, 2020.